Carriers which transport a single substrate through a membrane down its concentration gradient are called uniporters. Many carriers transport substrates against or up concentration gradients also. In such cases, the carrier transport across the membrane is combined with the transport of one more substrate down its electrochemical gradient e.g. Glucose. Its movement from plasma to red blood cells (down the concentration gradient) and to intestines (up its concentration gradient) is along with Na^+. It is called as symporter, as both glucose and Na^+ are transferred in the same direction.

Another class of carriers are termed as antiporters. These are called anti as they transport substrates in the opposite directions to the ion gradient promoting the reaction. All of the carrier-mediated responses are reversible.

It is well understood that the membrane effectively protects the interior of the cell from the exterior. Though the membrane is generally impermeable, molecules from the outside environment are taken up via a process called endocytosis

1.3.4.Endocytosis

In the process of endocytosis, an area or a region of the membrane would surround the substance which needs to be internalised. The section of the membrane that has surrounded the material would bud into the cell forming a vesicle which retains the ingested substance.

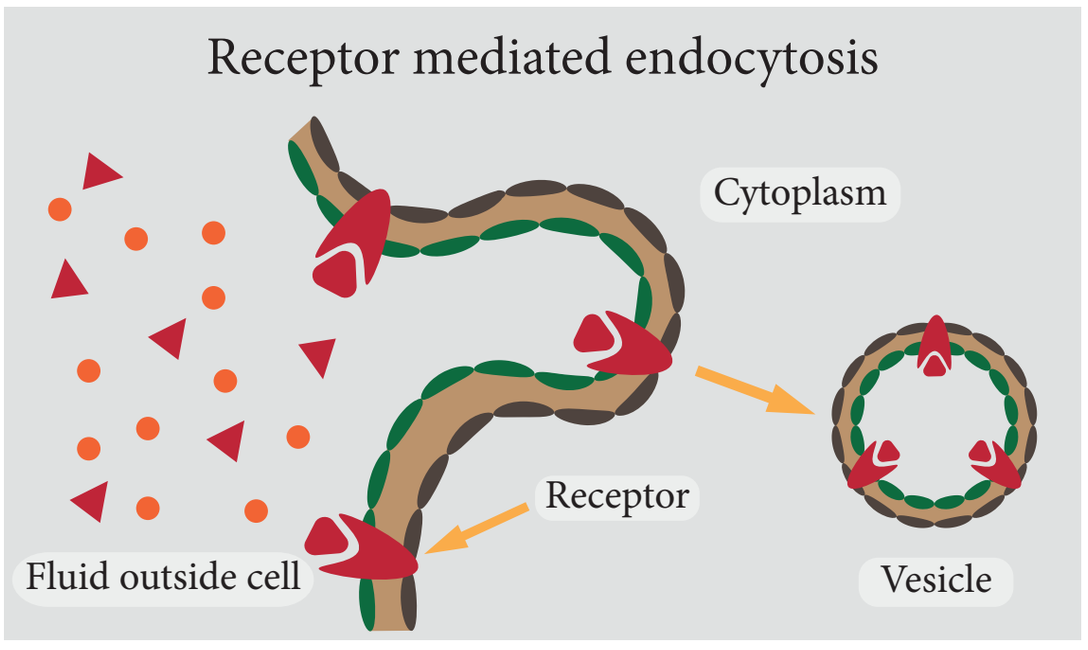


Figure7 .Receptor-mediated endocytosis. A particular substance binds to a receptor on the outer surface of the membrane and uptaken by endocytosis

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